

Assignment 3 – Shell Script to Snakemake, S3 Integration, and GitHub

This lab guides you through converting a public genomic analysis shell script from GitHub into a reproducible Snakemake workflow, integrating S3 cloud storage, and tracking your progress with your own GitHub repository.

Part A: Download Shell Script Pipeline from GitHub

1. Open your BIN5506 EC2 Class Server in VSCode.
2. Open the terminal and activate your Anaconda environment with all required dependencies.
3. Download the provided folder from the GitHub repository
 - o git clone https://github.com/SohailHanif/Bioinformatics_Cloud_Computing
4. Verify that the folder contains the genomic analysis pipeline shell script
5. Screenshot Requirements:
 - o VSCode view with the open pipeline shell script file
 - o Terminal view displaying the pipeline shell script executing successfully.

Part B: Set Up Your Own GitHub Repository

1. Using GitHub, create a new private repository named FIRSTNAME_assignment_2 (replace FIRSTNAME with your own first name).
2. Initialize a Git repo in your working directory, add your GitHub repo as a remote, and configure user settings.
3. Create an empty Snakemake File (Snakefile)
4. Commit and push file, for every rule in the Snakemake conversion, create a commit to show the new rule
5. Screenshot Requirements:
 - o GitHub repository page showing your empty Snakemake file
 - o Terminal showing git commit and git push commands

Part C: Convert Shell Script to Snakemake Workflow

1. Break down each major analysis step in the bash script (e.g., data download, alignment, sorting, variant calling).
2. Rewrite each step as a Snakemake rule, preserving the logical workflow and dependencies.
3. Test execution of each step as you work on it.
4. Once a step is working correctly create a git commit and push for that step.
5. Screenshot Requirements:
 - o Each step in the editor and its execution in the terminal

- Terminal showing git commit and git push commands
- Git log command showing all rule commits
- GitHub showing all rule commits

Part D: Create and Configure a New S3 Bucket

1. Create an S3 Bucket named FIRSTNAME_assignment_2.
2. Screenshot Requirements
 - New empty S3 bucket named FIRSTNAME_assignment_2.

Part E: Integrate S3 Upload Functionality

1. Add Snakemake Rule for upload as final rule
2. Get AWS credentials from learner lab start page
3. Paste credentials in your ~/.aws/credentials file (will expire after certain amount of time)
4. Execute and verify all files successfully uploaded to S3
5. Screenshot Requirement:
 - Snakemake S3 upload rule
 - Git commit S3 upload rule
 - AWS credentials in learner lab
 - AWS credentials in ~/.aws/credentials file
 - S3 bucket contents visible in the AWS Console

Submission

Submit a PDF with all screenshots labelled with the question number.